

Agroforestry

Spatial scale: farm

About this sustainable management strategy

Agroforestry combines grape production with trees and shrubs to create more diverse and multifunctional farming systems. Rather than separating productive and ecological areas, woody vegetation is intentionally integrated into the vineyard to strengthen ecosystem functions while providing additional environmental and economic benefits. Depending on the site and management objectives, agroforestry can include tree rows between vineyard blocks, fruit or timber trees along field margins, scattered trees within the vineyard, or wooded buffer zones. The selection, arrangement, and density of woody vegetation should be adapted to local soil conditions, water availability, topography, and vineyard operations.

The presence of trees and shrubs contributes to healthier and more resilient vineyard ecosystems. Their root systems improve soil structure and nutrient cycling, while the canopy moderates wind speed, temperature, and solar radiation. Woody vegetation also creates habitats and ecological corridors for pollinators, birds, and natural enemies of pests. In addition to supporting grape production, agroforestry can diversify farm outputs through products such as fruit, timber, biomass, fodder, or other non-timber products, strengthening the long-term sustainability of the farming system.

Key ecosystem services

- Soil health and fertility (*improved soil structure, nutrient cycling, and organic matter accumulation*)
- Water regulation (*enhanced infiltration, reduced runoff, and improved water retention*)
- Erosion control (*soil stabilisation through root systems*)
- Microclimate regulation (*wind buffering, shading, and temperature moderation*)
- Habitat provision and biodiversity conservation (*food resources, shelter, and ecological connectivity*)
- Natural pest control (*support for beneficial insects, birds, and other natural enemies*)
- Cultural heritage and landscape aesthetics (*maintenance of traditional multifunctional landscapes*)

Considerations for successful implementation

Spatial design and vineyard operations

The arrangement of trees and shrubs should maintain efficient access for machinery, pruning, spraying, and harvesting while avoiding excessive shading of vines. Tree spacing, orientation, and expected canopy development should be considered during the design phase.

Long-term establishment and management

Agroforestry systems develop over many years and require regular management during establishment, including pruning, weed control, and protection of young trees. Periodic maintenance helps maintain a balance between tree growth and vineyard productivity.

Management of resource interactions

Trees and vines interact through their use of water, nutrients, and light. Species selection, planting density, and appropriate spacing can help maximise complementary interactions while reducing competition, particularly in dry environments or on shallow soils.